Upregulation of lymphocyte apoptosis as a strategy for preventing and treating autoimmune disorders: a role for whole-food vegan diets, fish oil an...

Induced apoptosis of autoreactive T-lymphocyte precursors in the thymus is crucial for the prevention of autoimmune disorders. IGF-I and prolactin, which are lymphocyte growth factors, may have the potential to suppress apoptosis in thymocytes and thus encourage autoimmunity; conversely, dietary fish oil rich in omega-3 fats appears to upregulate apoptosis in lymphocytes. Since whole-food vegan diets may downregulate systemic IGF-I activity, it is proposed that such a diet, in conjunction with fish oil supplementation and treatment with dopamine agonists capable of suppressing prolactin secretion, may have utility for treating and preventing autoimmune disorders. This prediction is consistent with the extreme rarity of autoimmune disorders among sub-Saharan black Africans as long as they followed their traditional quasi-vegan lifestyles, and with recent ecologic studies correlating risks for IDDM and for multiple sclerosis mortality with animal product and/or saturated fat consumption. Moreover, there is evidence that vegan or quasi-vegan diets are useful in the management of rheumatoid arthritis, multiple sclerosis, and possibly SLE. The dopamine agonist bromocryptine exerts anti-inflammatory effects in rodent models of autoimmunity, and there is preliminary evidence that this drug may be clinically useful in several human autoimmune diseases; better tolerated D2-specific agonists such as cabergoline may prove to be more practical for use in therapy. The moderate clinical utility of supplemental fish oil in rheumatoid arthritis and certain other autoimmune disorders is documented. It is not unlikely that extra-thymic anti-inflammatory effects contribute importantly to the clinical utility of vegan diets, bromocryptine, and fish oil in autoimmunity. The favorable impact of low latitude or high altitude on autoimmune risk may be mediated by superior vitamin D status, which is associated with decreased secretion of parathyroid hormone; there are theoretical grounds for suspecting that parathyroid hormone may inhibit apoptosis in thymocytes. Androgens appear to up-regulate thymocyte apoptosis, may be largely responsible for the relative protection from autoimmunity enjoyed by men, and merit further evaluation for the management of autoimmunity in women. It will probably prove more practical to prevent autoimmune disorders than to reverse them once established; a whole-food vegan diet, coupled with fish oil and vitamin D supplementation, may represent a practical strategy for achieving this prevention, while concurrently lowering risk for many other life-threatening 'Western' diseases.